Applied Electronics

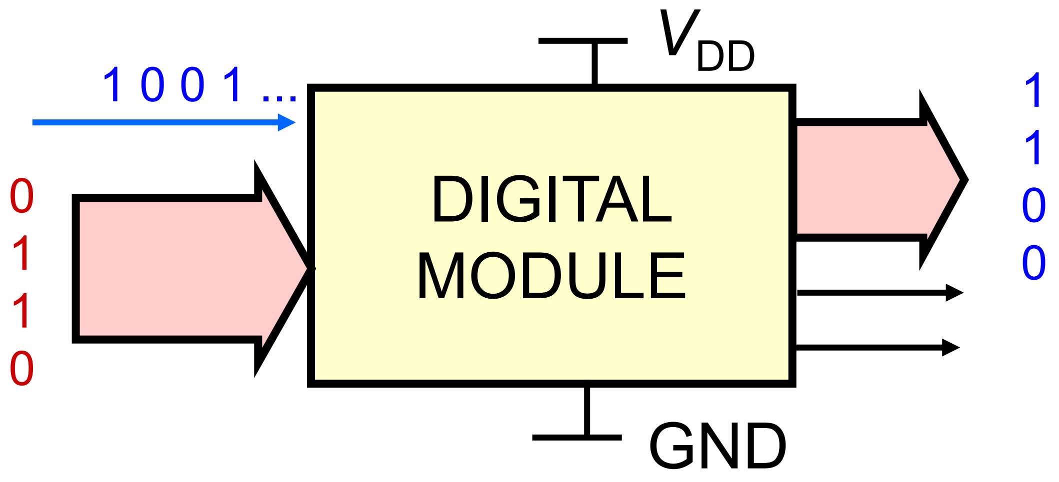
Combinational Circuits

Logic Circuits

- Static electrical parameters
  - Logic states and levels
  - Output characteristics, interfacing
- Output circuits
  - Totem Pole, Open Collector, Three State
- Power dissipation
- Dynamic parameters
  - Definition, Models
- Logic families

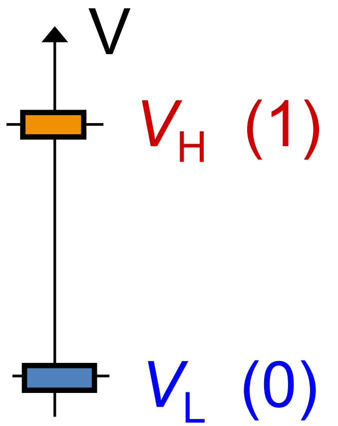
Digital Modules

- A digital module uses
  - ♦ A Power Supply (V_{DD}, GND)
  - ♦ Input and out signals, represented as binary words
    - Groups of logical values 1/0, in serial or parallel format



Logic States and Voltage Levels

- Logic states (0/1, L/H) are represented by electrical quantities (very often voltages, V_H, V_L)
- Assignment logic state ↔ voltage is arbitrary
  - "positive logic": $1 \leftrightarrow V_H$, $0 \leftrightarrow V_L$ ➔ used in this course
  - "negative logic": $0 \leftrightarrow V_H$, $1 \leftrightarrow V_L$
- In this course
  - State 1, H: highest voltage, V_H, slightly less than V_{cc}
  - State 0, L: lowest voltage, V_L, slightly higher than GND (0 V)

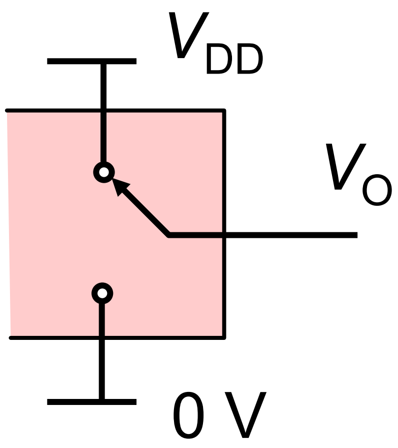


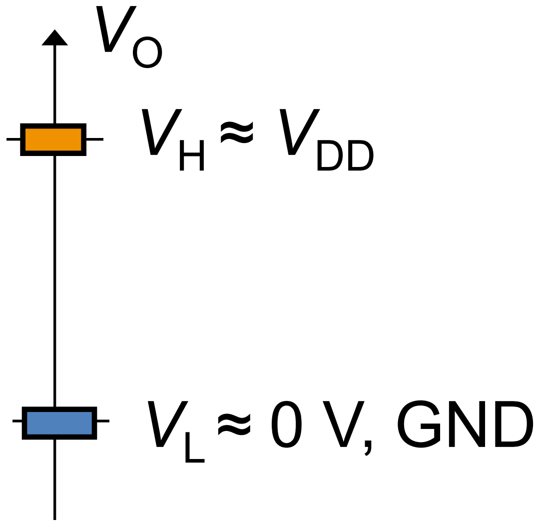
Logic States and Voltage Levels

- V_H and V_L obtained from power supply, V_{DD}, and the 0 V reference (GND), respectively
- As first approximation $V_H = V_{DD}$ and $V_L = 0$ V

state 1, H:
$V_O = V_H$

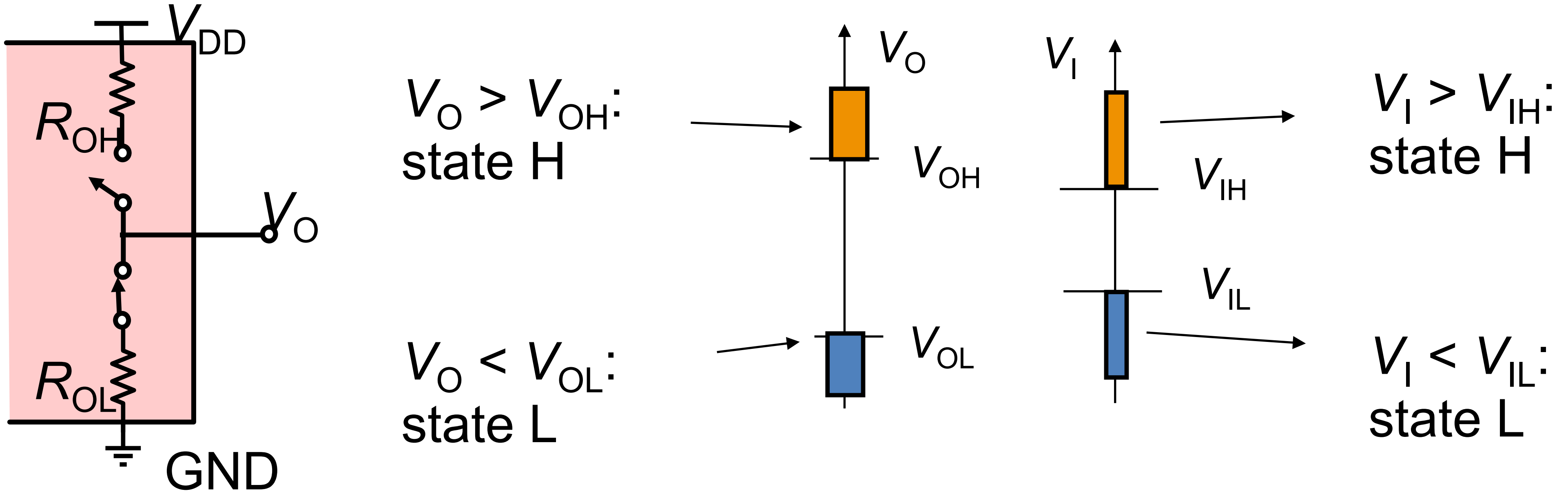
state 0, L:
$V_O = V_L$





Voltage Range for H and L Levels

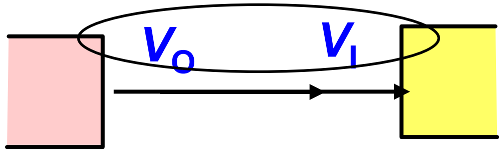
- Output final voltage depends on: R_{out}, on the load, on the V_{DD}
  - We define: V_{OH} and V_{OL}
  - We define: V_{IH} and V_{IL}

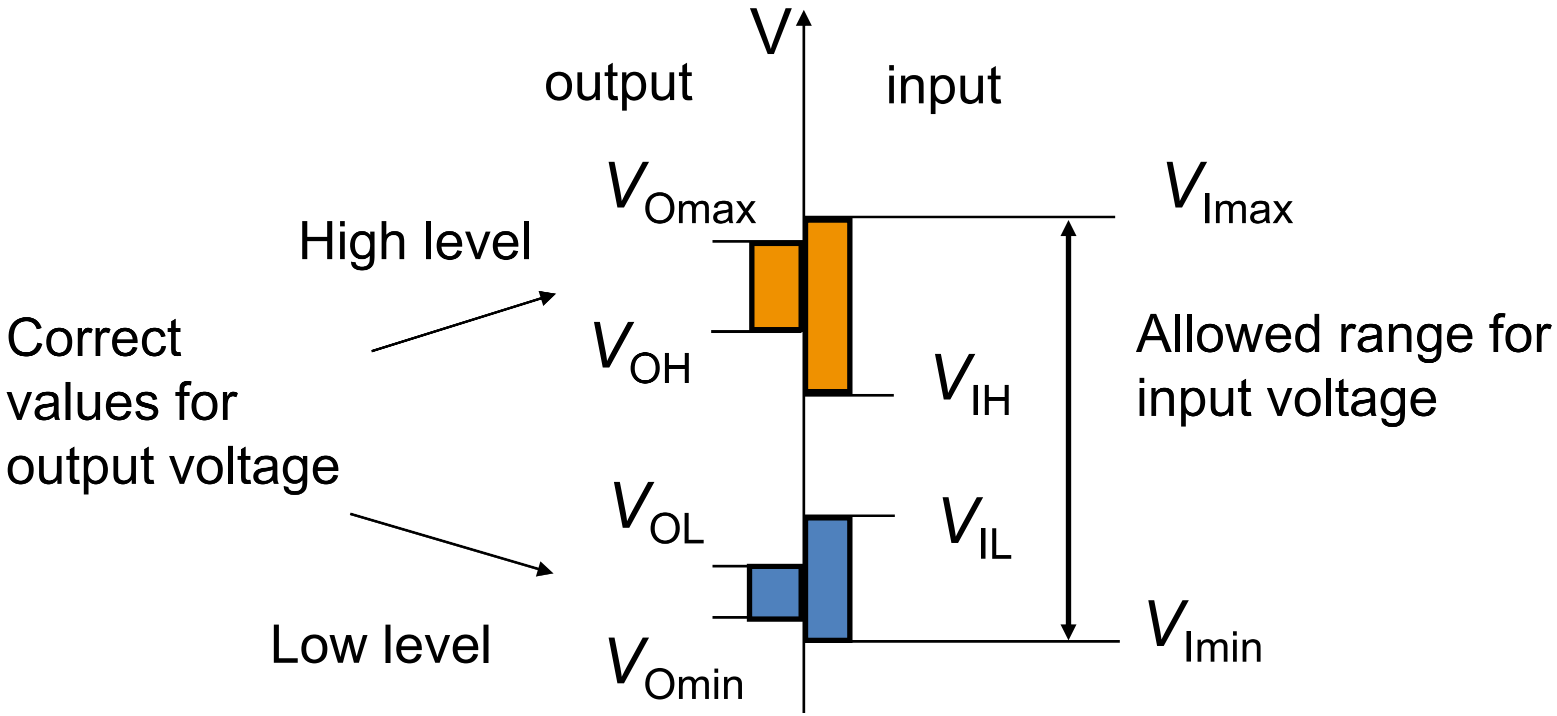


Input and Output Voltage Range

- Voltage RANGES, not levels!

The output of one logic circuits is the input of the next one



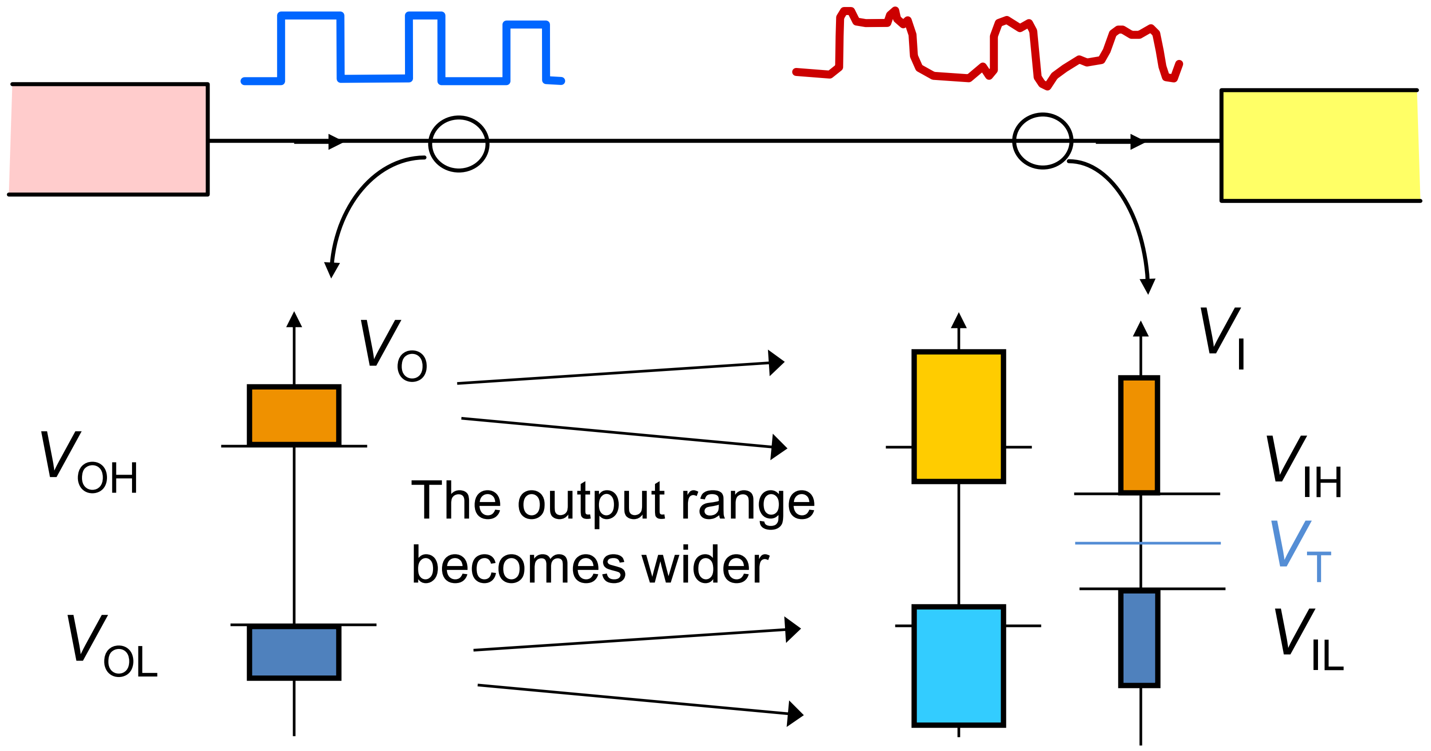


Compatibility of Logic Circuits

- Compatible logic circuits can transfer logic states
  - Inputs detect properly the output voltage levels
- Conditions for logic compatibility
  - $V_{OL} < V_{IL}$
    - Guarantees that the output voltage for the low logic state is read as low state by the driven logic inputs
  - $V_{OH} > V_{IH}$
    - Guarantees that the output voltage for the high logic state is read as high state by the driven logic inputs

Actual Signals

- Noise and interferences modify the output voltage

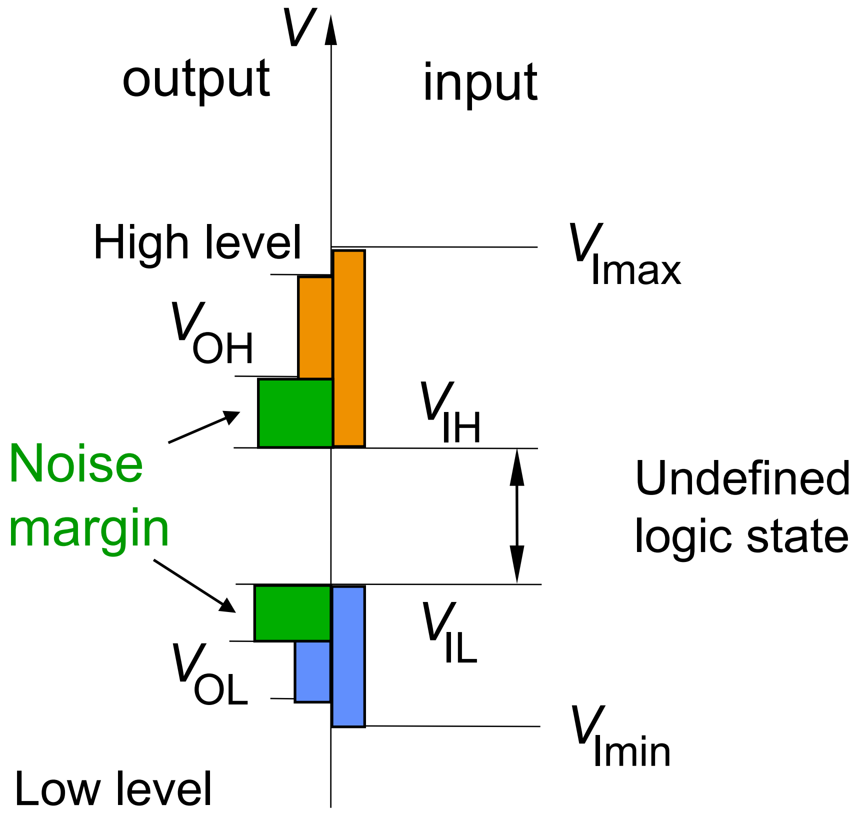


Noise Margins

- The gaps $V_{OH} - V_{IH}$ and $V_{OL} - V_{IL}$ guarantee the correct sensing of the logic states, even with noise

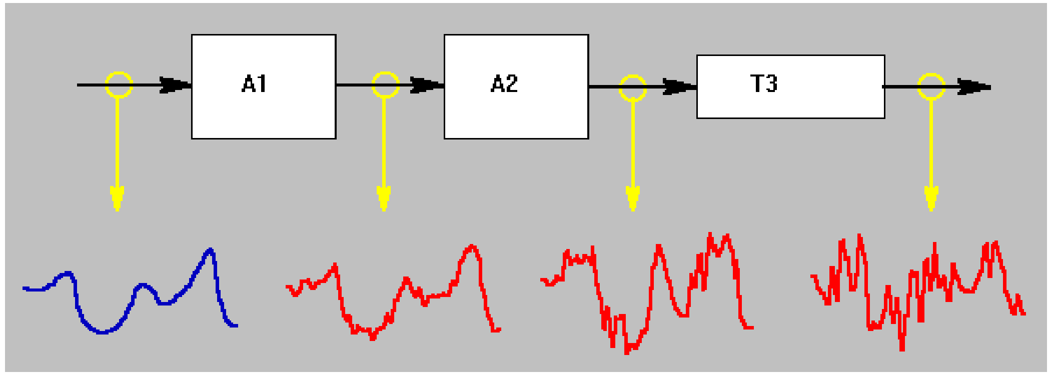
- The gap is defined as noise margin:

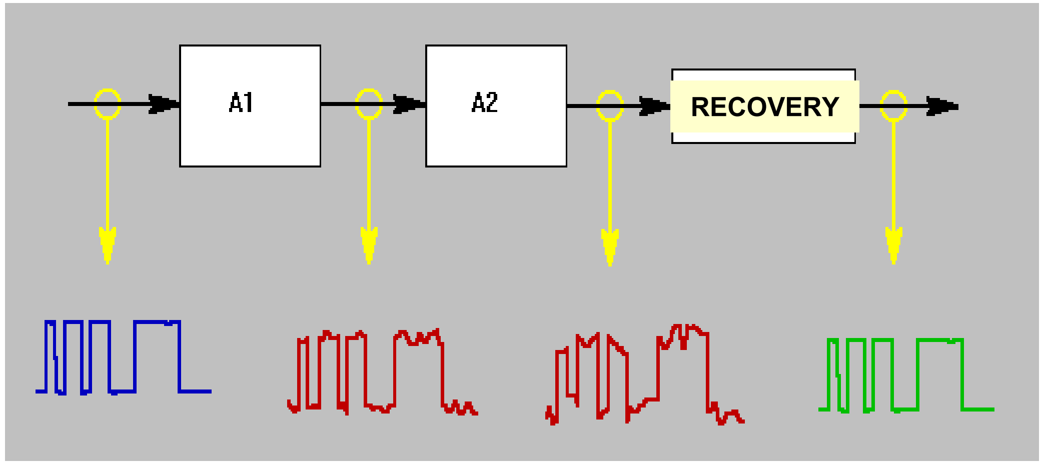
$$NM_H = V_{OH} - V_{IH}$$
$$NM_L = V_{IL} - V_{OL}$$



Noise on Digital & Analog Signals

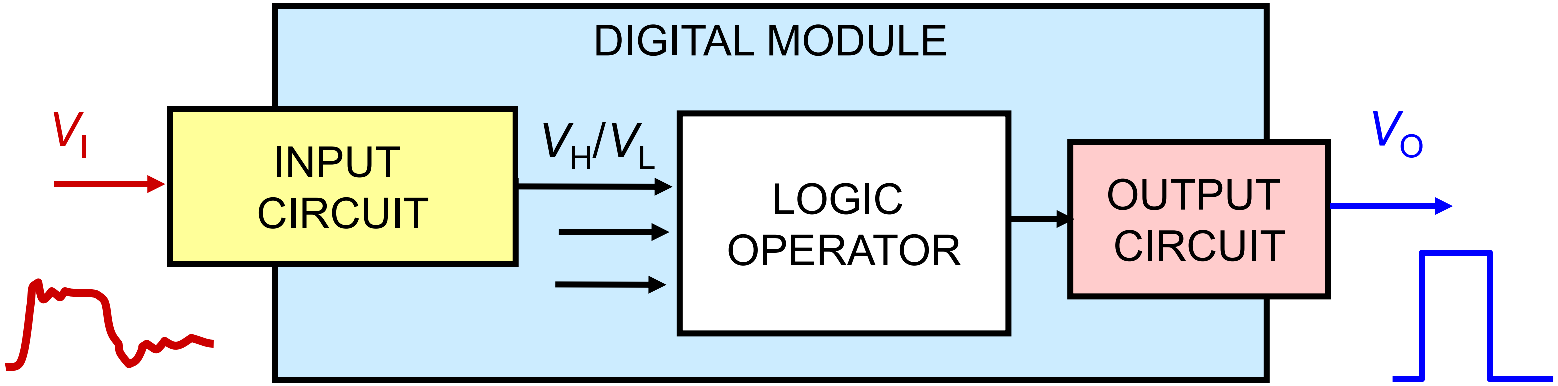
- Any processing step adds noise
- Analog signal
  - Noise means *unrecoverable* information loss
- Digital signal
  - Signal degradation can be recovered within the noise margin boundaries



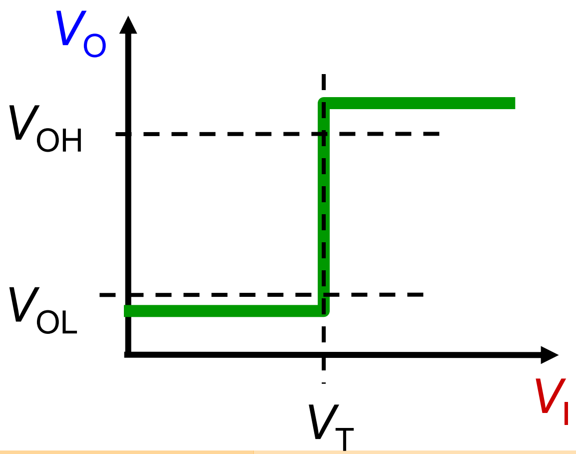


Digital Signal Recovery

- All digital inputs compare V_I with a threshold

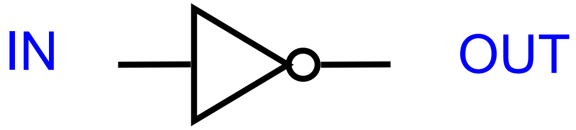


- The input circuit compares V_I against V_T
- The output circuit generates V_O (with noise may exceed V_{OH}, V_{OL})



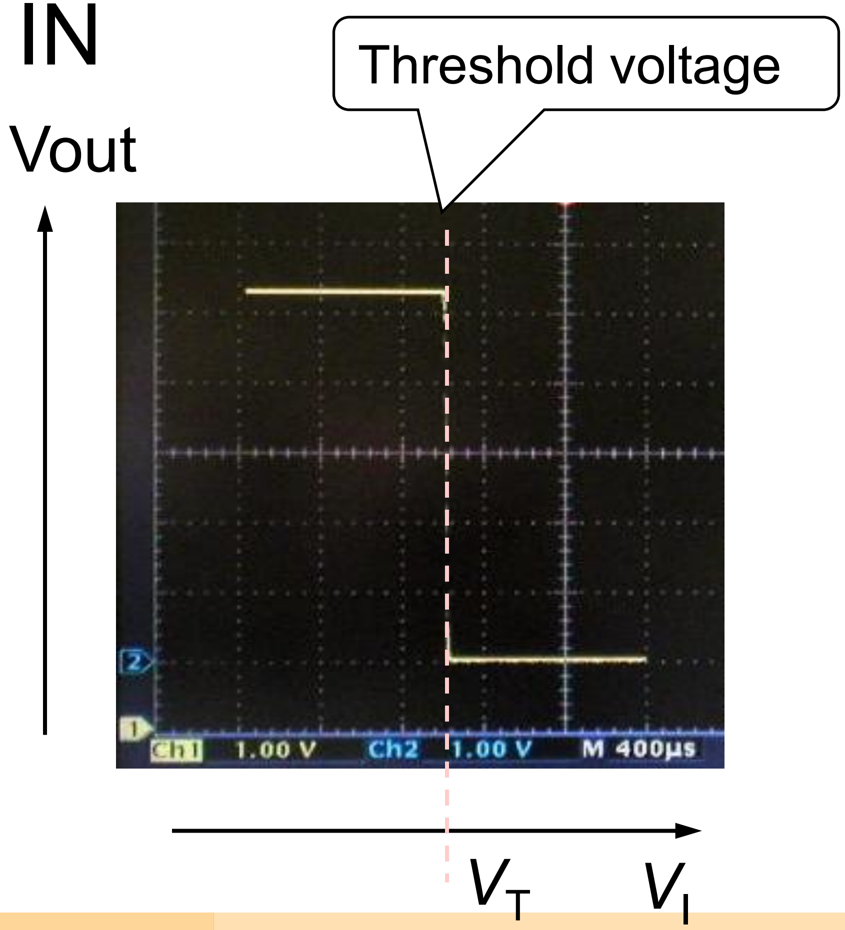
Example: Logic Inverter

- Schematic symbol



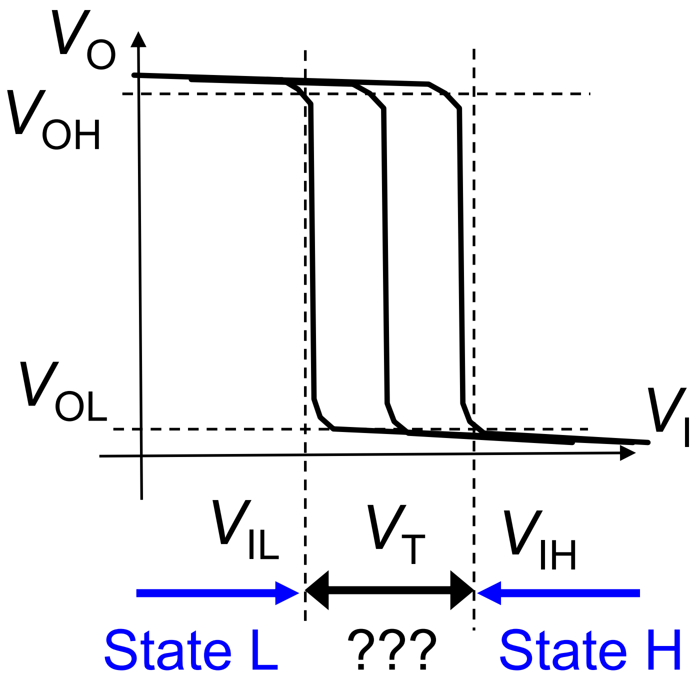
- Logic operation: OUT = NOT IN

- Transfer characteristic
  - ♦ On X axis: the allowed values of the input voltage
  - ♦ On Y axis: the measured value of the output voltage for each V_I



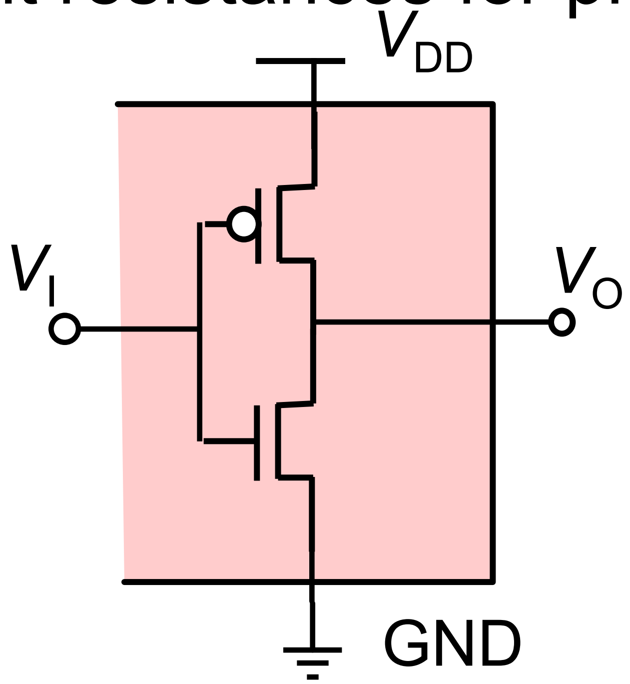
Definition of V_{IH} and V_{IL}

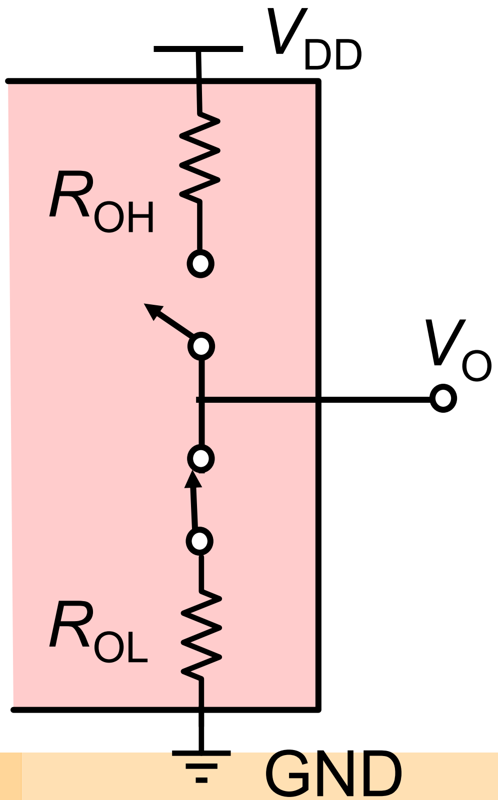
- V_T can change with the power supply voltage, temperature, ….
  - We can only define a voltage range for V_T
- $V_I > V_{IH}$ → Logic state High
- $V_I < V_{IL}$ → Logic state Low
- $V_{IL} < V_I < V_{IH}$ →
  - Logic state detection *undefined*



Output Circuit of a CMOS Gate

- The output nMOS and pMOS are modeled with their **equivalent resistance *R_O***
  - Connect the output towards GND or V_{DD}
  - Different resistances for pMOS and nMOS





Output Currents

- V_{OL} and V_{OH} guaranteed if output current I_O is within a specific range
  - $I_{OL} < I_{OLmax}$ guarantees V_{OL}
  - $I_{OH} < I_{OHmax}$ guarantees V_{OH}
- I_O is determined by voltage level and load(s)
  - Input currents of driven gates (CMOS → negligible)
  - Input currents of loads (LED, resistances, ...)

Static Logic Compatibility Check

1) Separate the analysis for the low and high states
2) Check voltage level compatibility
   - $V_{OL} < V_{IL}$; $V_{OH} > V_{IH}$
3) Calculate output currents for low and high states
   - Add input currents of connected logic circuits I_{IL}, I_{IH}
   - Load currents
     - Low state: loads connected to V_{DD}
     - High state: loads connected to GND
4) Check current compatibility
   - Low state: $|I_O| < |I_{OL}|$; high state: $|I_O| < I_{OH}$

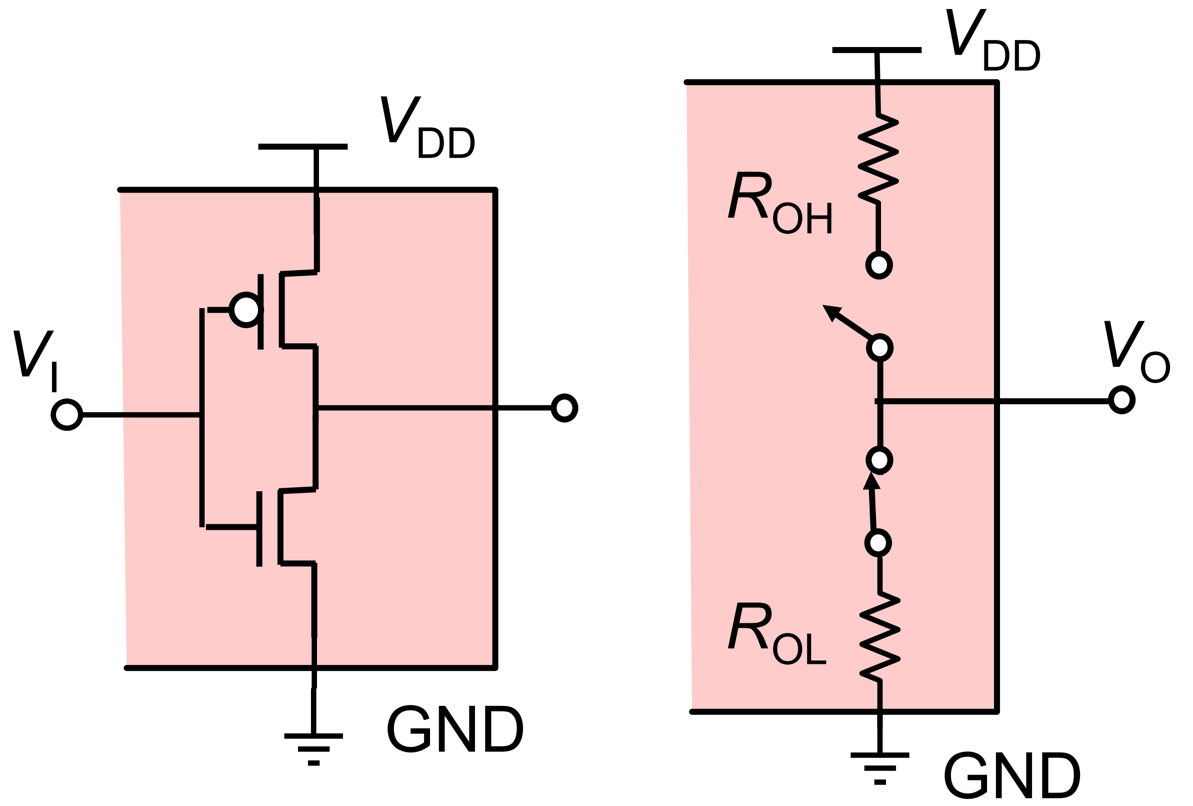
Output Electrical Parameters

- Data-sheet of gates
  - ♦ Voltages V_{OL}, V_{OH}, and Currents I_{OL}, I_{OH}

PARAMETER	TEST CONDITIONS		V_{CC}	$T_A = 25°C$ MIN	TYP	MAX	UNIT
V_{OH}	$V_I = V_{IH}$ or V_{IL}	$I_{OH} = -20\ \mu A$	2 V	1.9	1.998		V
			4.5 V	4.4	4.499		
			6 V	5.9	5.999		
		$I_{OH} = -4$ mA	4.5 V	3.98	4.3		
		$I_{OH} = -5.2$ mA	6 V	5.48	5.8		
V_{OL}	$V_I = V_{IH}$ or V_{IL}	$I_{OL} = 20\ \mu A$	2 V		0.002	0.1	V
			4.5 V		0.001	0.1	
			6 V		0.001	0.1	
		$I_{OL} = 4$ mA	4.5 V		0.17	0.26	
		$I_{OL} = 5.2$ mA	6 V		0.15	0.26	

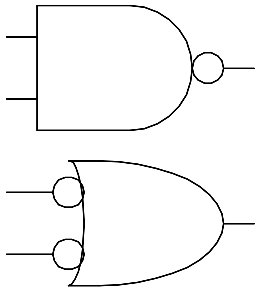
Totem Pole Output Circuit

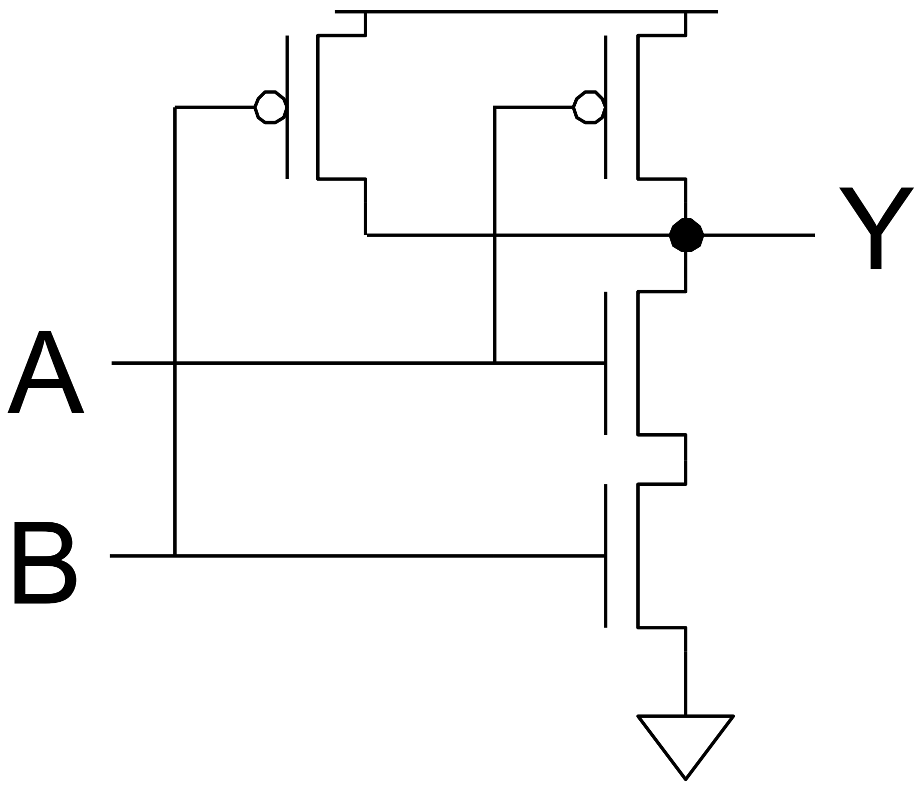
- Model the output as a switch connected to V_{DD} (logic state H) or to GND (logic state L)
- State H:
  output voltage close to V_{DD}
- State L:
  output voltage close to GND
- All CMOS gates are *inverting* (NAND instead of AND, NOR instead of OR, …)
  - Why?



Two-Input CMOS NAND

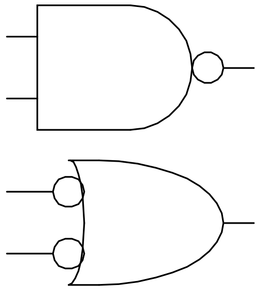
A	B	Y
0	0	
0	1	
1	0	
1	1	

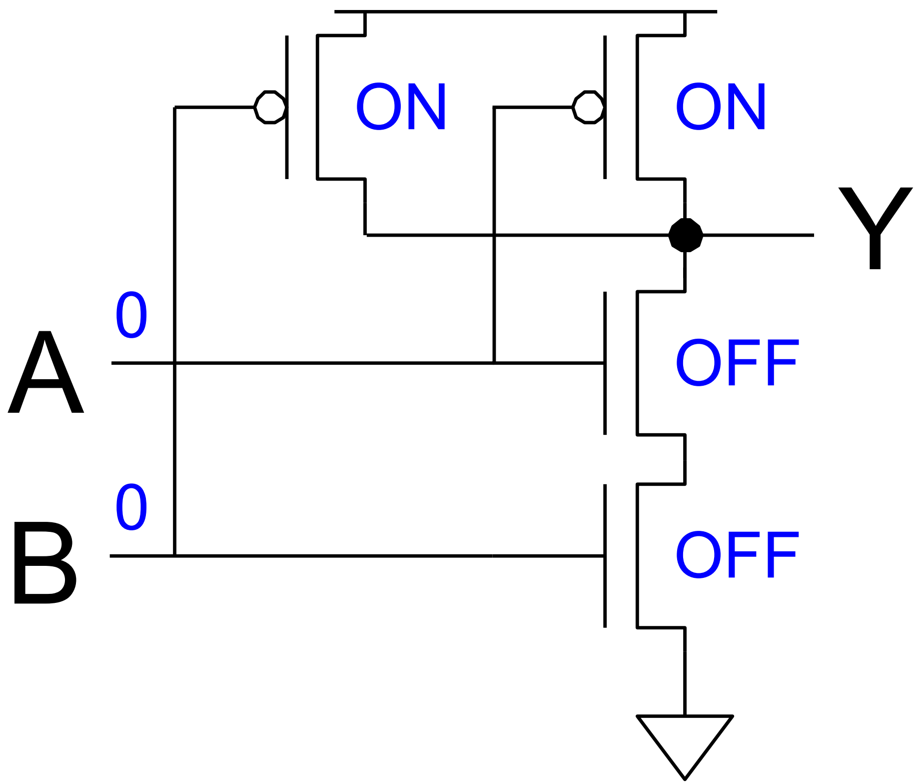




Two-Input CMOS NAND

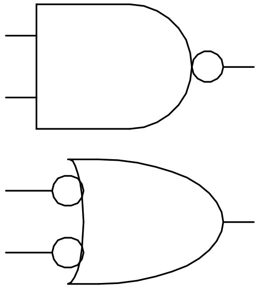
A	B	Y
0	0	1
0	1	
1	0	
1	1	

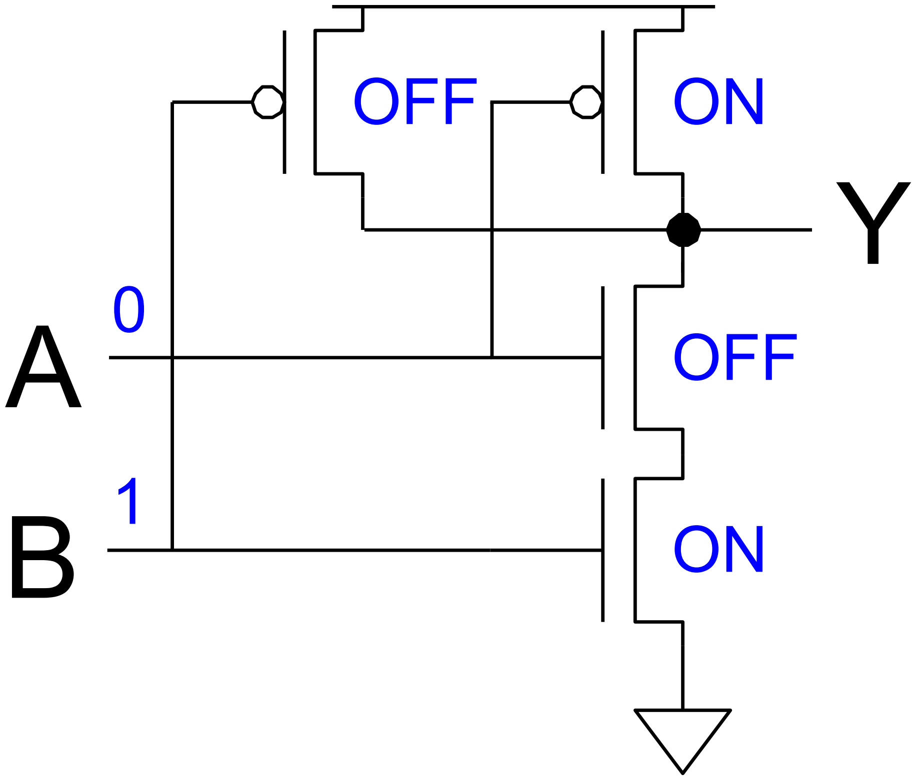




Two-Input CMOS NAND

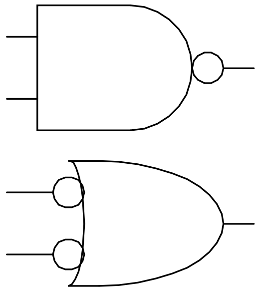
A	B	Y
0	0	1
0	1	1
1	0	
1	1	

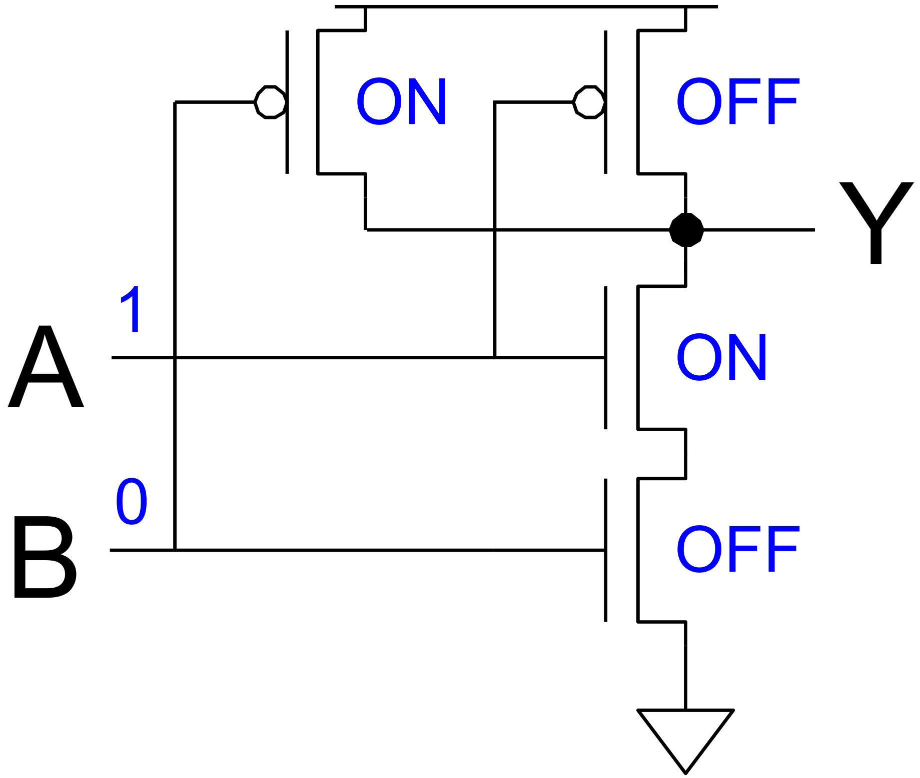




Two-Input CMOS NAND

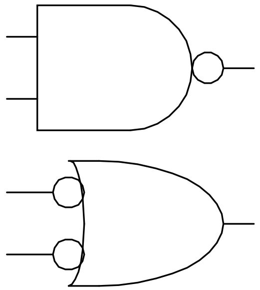
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	

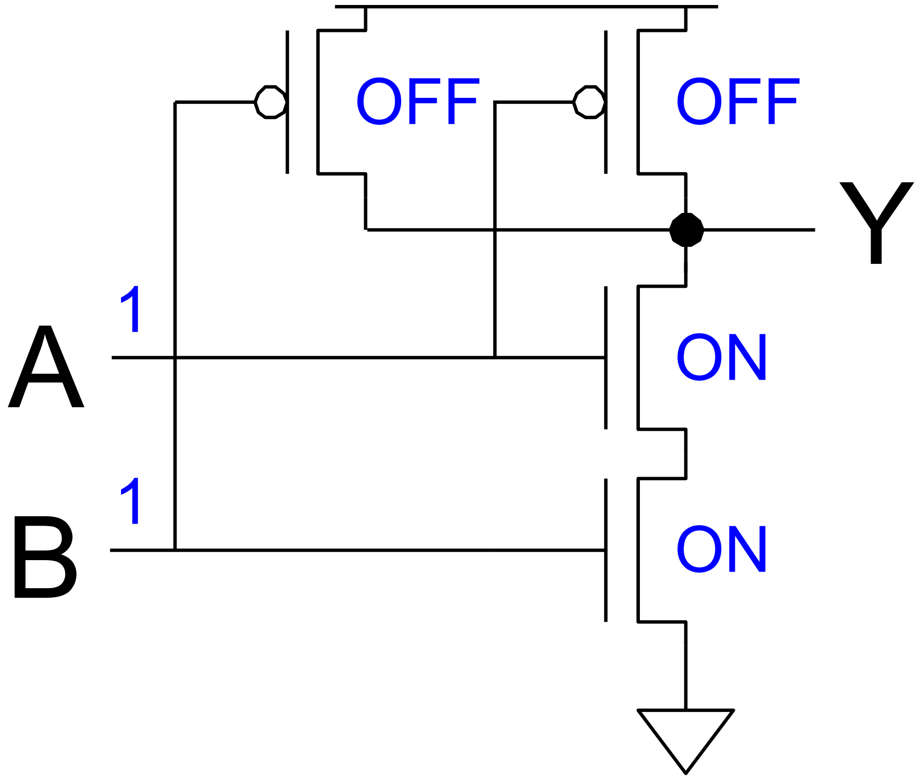




Two-Input CMOS NAND

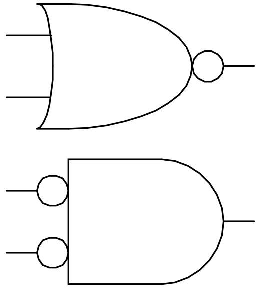
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

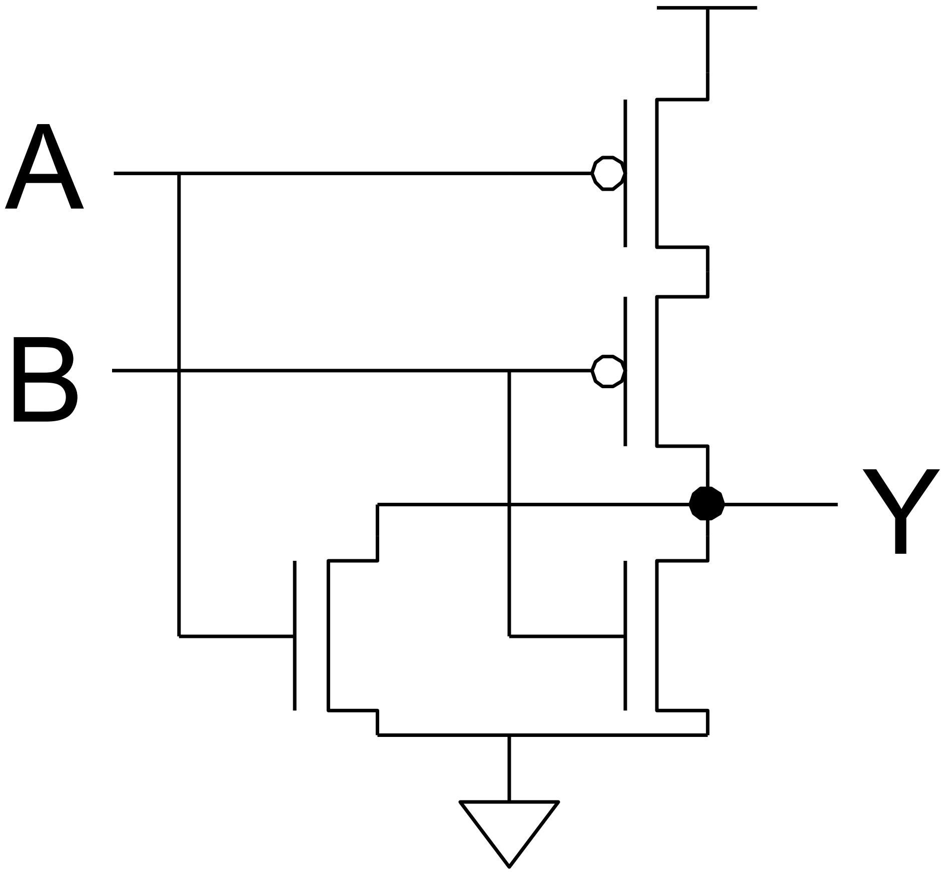




Two-Input CMOS NOR

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0





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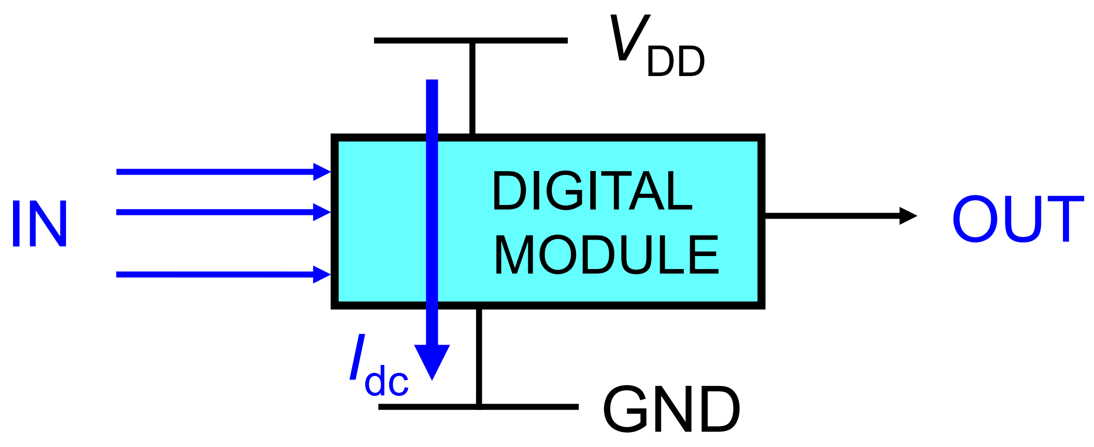
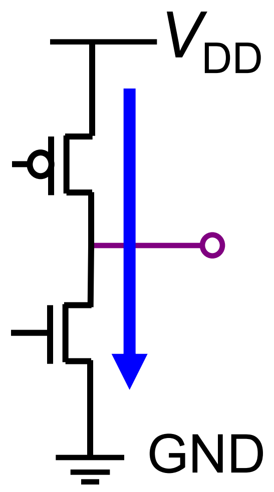
Static Power P_S

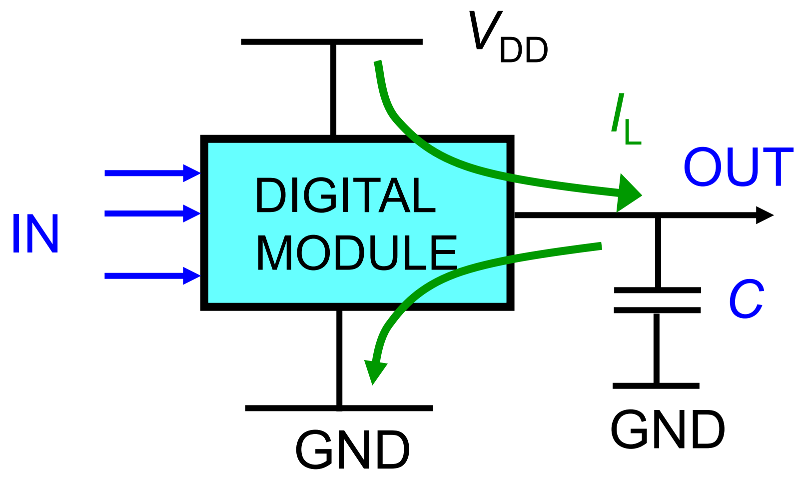
- Power consumed without switching
  - Roughly constant
  - Depends on temperature
  - Model: continuous current (I_{dc}) between V_{DD} and GND

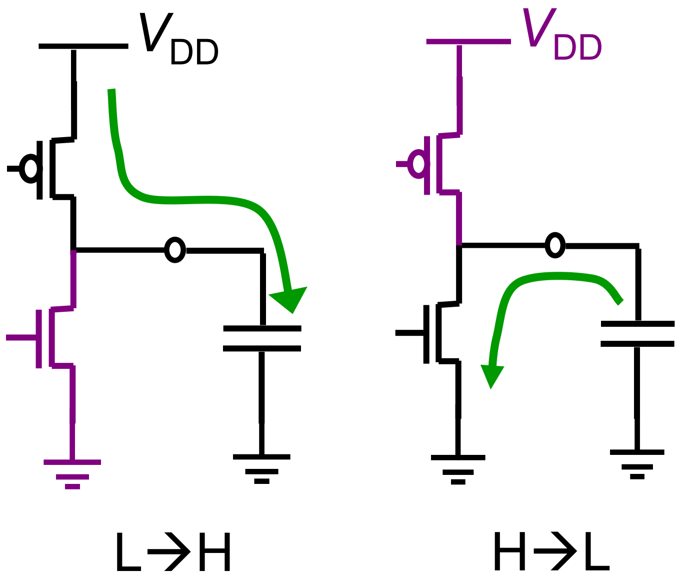




Dynamic Power P_D

- Power consumed when switching (H→L or L→H)
  - Current I_L that charges/discharges output capacitance
  - Impulsive, only during switching





Calculation of Dynamic Power P_D

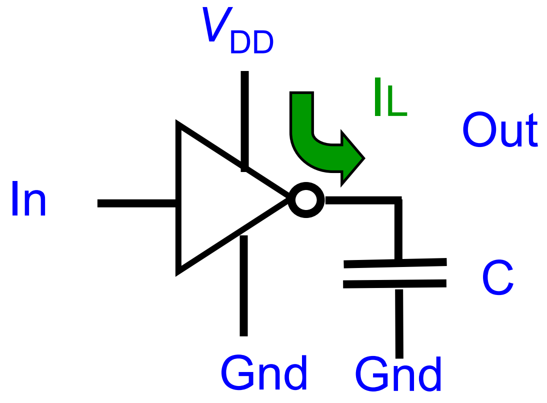
- CMOS gates: almost no static current
- Outputs mostly charge-discharge capacitances
  - ♦ On H→L, all energy in C is dissipated by R_ON of nMOS

$$\mathcal{E}_\mathrm{T} = \frac{1}{2} C V^2$$

  - ♦ On L→H R_ON of pMOS dissipates the same energy
- Charge/discharge C for f times/s

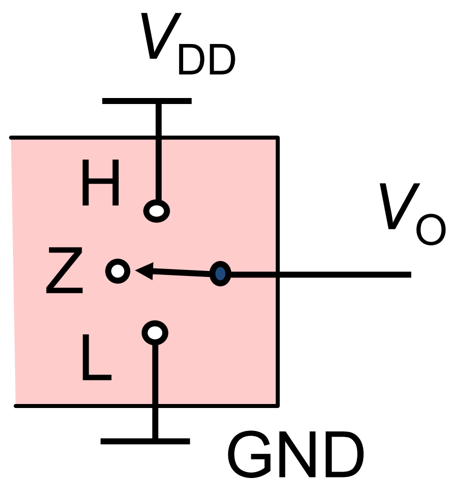
$$P_\mathrm{D} = \frac{2\mathcal{E}_\mathrm{T}}{\mathrm{T}} = 2\left(\frac{1}{2} C V^2\right) f = f C V^2$$

  - ♦ Depends on V, C, f (and technology)



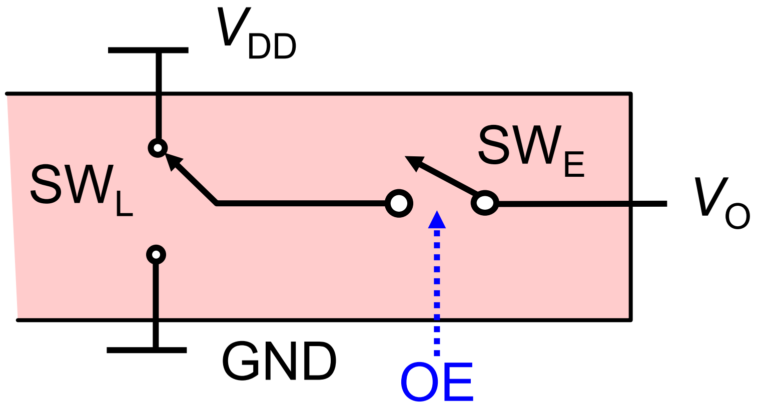
Three-State Output Circuit

- Three different output states
  - ♦ H: logic level HIGH (1)
  - ♦ L: logic level LOW (0)
  - ♦ Z: high impedance
    - Neither switch (to V_{DD} or V_{GND}) is closed
- We need two control signals
  - ♦ One signal to set the H/L states
  - ♦ One signal (ENABLE) to set the Z state
    - ENABLE = 0 forces Z state



Enable Model for a 3-S Output

- Switch SW_L: single pole double through
  + Switch SW_E: single pole, single through



- SW_E is driven by OUT ENABLE (OE)
  - SW_L sets the logic state of the output (if OE = 1)
  - SW_E enables or disables the output

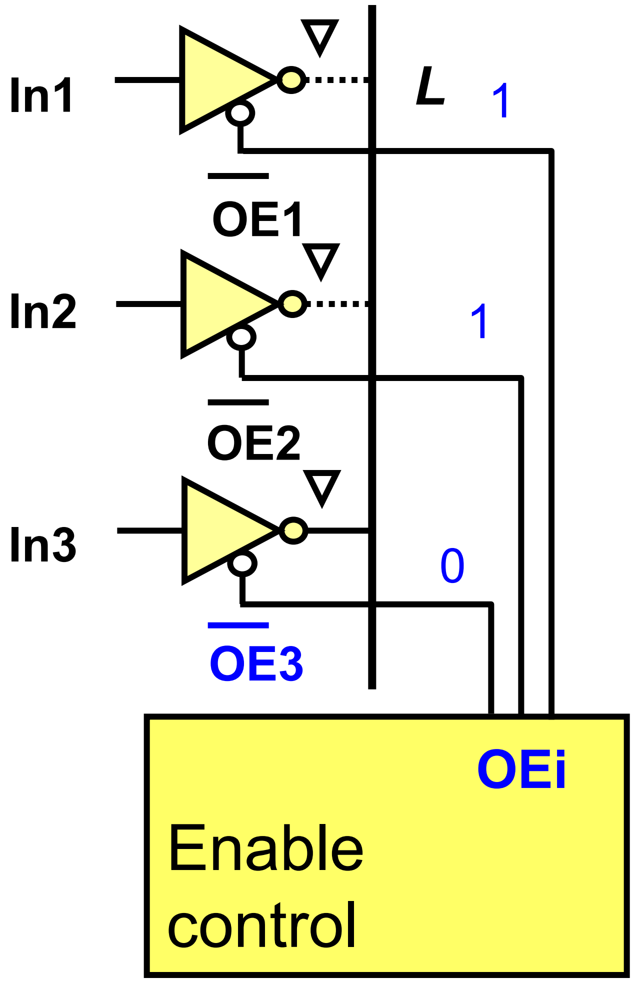
Output Parameters of a 3-S Output

- Enable = 1
  - Output like a totem pole with limits: voltage V_{OL}, V_{OH}, currents I_{OL}, I_{OH}
- Enable = 0 (HiZ, Open)
  - Leakage current: I_{OZ}
  - It is usually very small, much smaller than I_{OL}, I_{OH}
  - I_{OZ} comparable with CMOS input currents I_I

SYMBOL	PARAMETER	T_{amb} (°C) −40 to +85			UNIT	TEST CONDITIONS		
		MIN.	TYP.	MAX.		V_{CC} (V)	V_I	OTHER
I_{OZ}	3-state output OFF-state current	–	0.1	±10	µA	3.6	V_{IH} or V_{IL}	$V_O = V_{CC}$ or GND

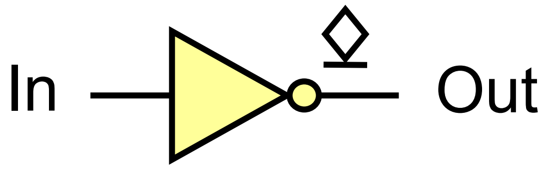
Application of Three-State Outputs

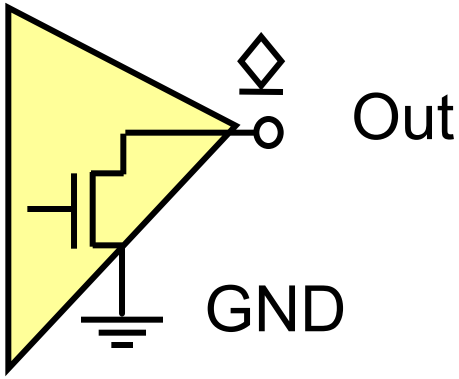
- A control unit issues mutually exclusive enable signals (enables one at a time)
  - “Knows” which to enable
- Application examples
  - Memory or register reading
  - Multiplexer
- Cannot be used if the pre-selection is not known
  - Example: interrupts



Open Collector (Open Drain) Out

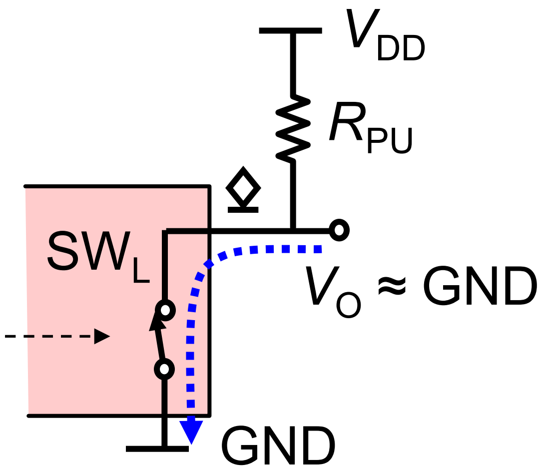
- Output has a single switch (MOS or BJT) to GND
- SW closed
  - ♦ GND (L) → Out
- SW open
  - ♦ Out disconnected (high Z)
    - Out state depends on external circuit
- No enable command
- No conflict
  - ♦ “Conflict” when the outputs of two TP gates drive *different* logic states on the same line



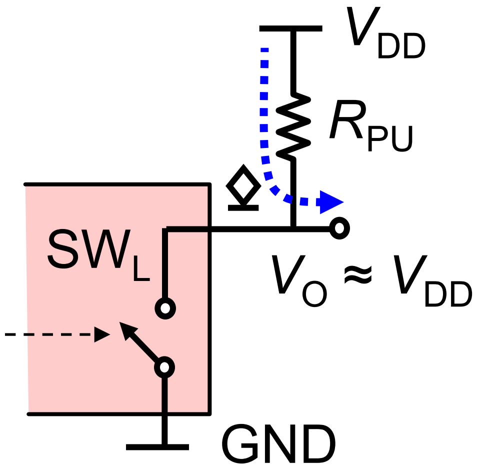


Open Collector/Drain Model

- The output MOS forces the low level by connecting the output to GND
  L → $V_O = 0$ V



- When the MOS is OFF, V_O goes to the high level thanks to an external pull-up resistor, R_{PU}
  H → $V_O = V_{DD}$



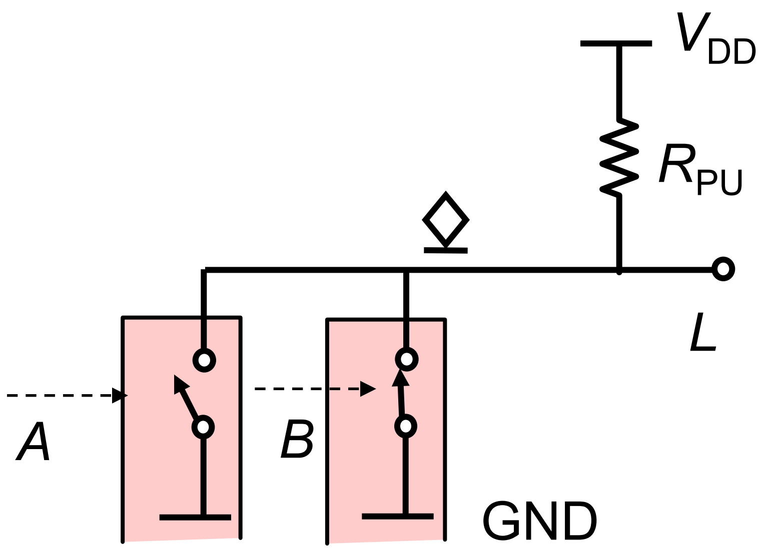
Params of Open Drain Gates

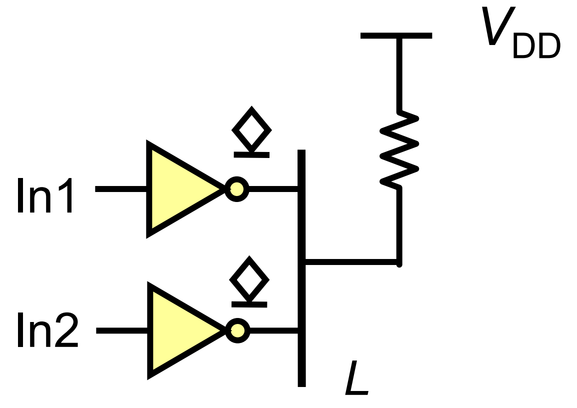
- Low state: same as totem pole output
  - Voltage: V_{OL}, current: I_{OL}
- High state (HiZ, open): same as a disabled 3-S
  - Leakage current, I_{OH}, very small, << I_O

PARAMETER	TEST CONDITIONS		V_{CC}	T_A = 25°C MIN	T_A = 25°C TYP	T_A = 25°C MAX	SN74HC05 MIN	SN74HC05 MAX	UNIT
I_{OH}	$V_I = V_{IH}$ or V_{IL},	$V_O = V_{CC}$	6 V		0.01	0.5		5	µA
V_{OL}	$V_I = V_{IH}$ or V_{IL}	I_{OL} = 20 µA	2 V		0.002	0.1		0.1	V
			4.5 V		0.001	0.1		0.1	
			6 V		0.001	0.1		0.1	
		I_{OL} = 4 mA	4.5 V		0.17	0.26		0.33	
		I_{OL} = 5.2 mA	6 V		0.15	0.26		0.33	

Application: Wired OR Operator

- Open Collector outputs can implement logic operators
  - ♦ L node is LOW if at least one of the A or B SWs is closed
  - ♦ Logic operator is often *NOR*
    - L = 0 when at least one of the inputs = 1 → L = NOT(A + B)
- But application is called "wired *OR*"
  - ♦ For an interrupt request line: In_1 *OR* In_2 can activate L





Logic With Open Collector Gates

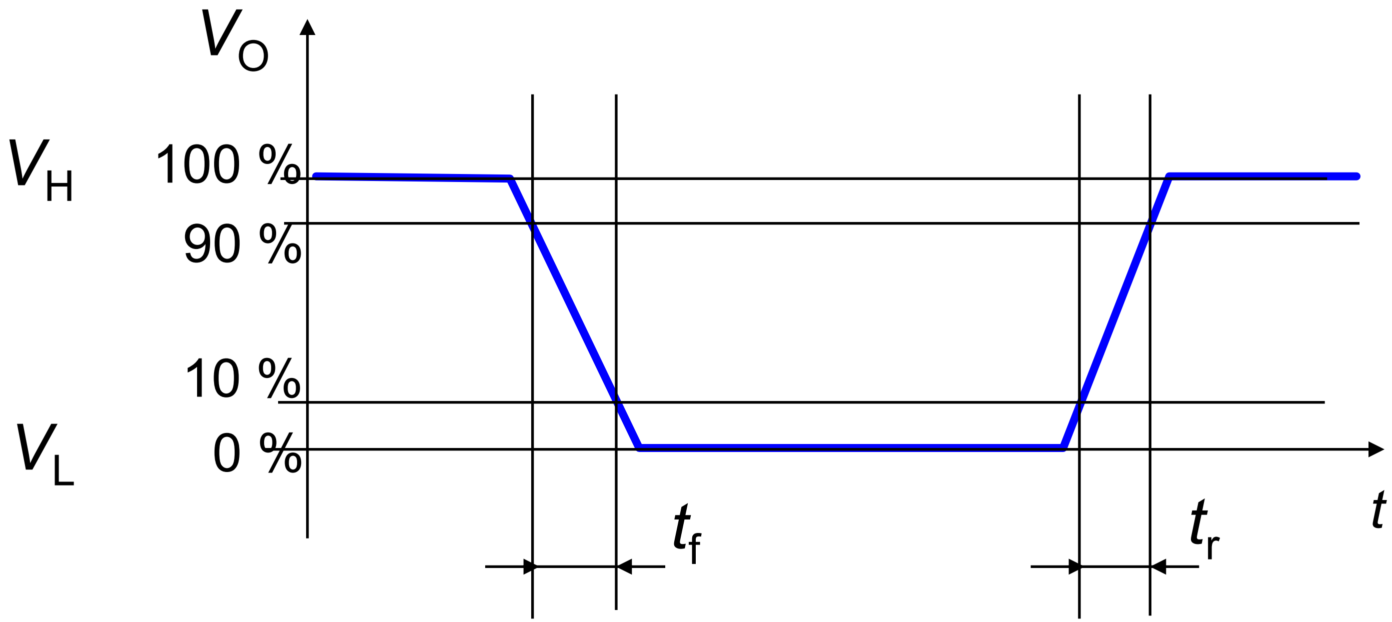
- By connecting several open collector gates to the same line/node (with pull-up resistors) we obtain
  - Wired OR (if L is active low)
    - Output node L is logic low if *at least* one output is logic low
      $L = \text{NOT}(A + B)$
  - Wired AND (if L is active high)
    - Output node L is logic high if *all* the outputs are logic high
      $L = \text{NOT}(A) \cdot \text{NOT}(B)$
- Can also change the number of circuits connected to a node, which is very useful in composable systems

Dynamic Parameters

- Input signals have finite slope edges
  - Input logic state change is recognized with a delay
- Variation of voltages and currents (due to input changes) is not instantaneous
  - *Delay* between input changes and output changes
- These effects cause
  - Delays of the propagation of the signals in the system
  - Limits of the circuit speed
    - Maximum switching frequency of the inputs
- Use a simplified RC linear model to analyze
  - Other models: LRC circuits, multiple cells, distributed parameters, non-linear devices ...

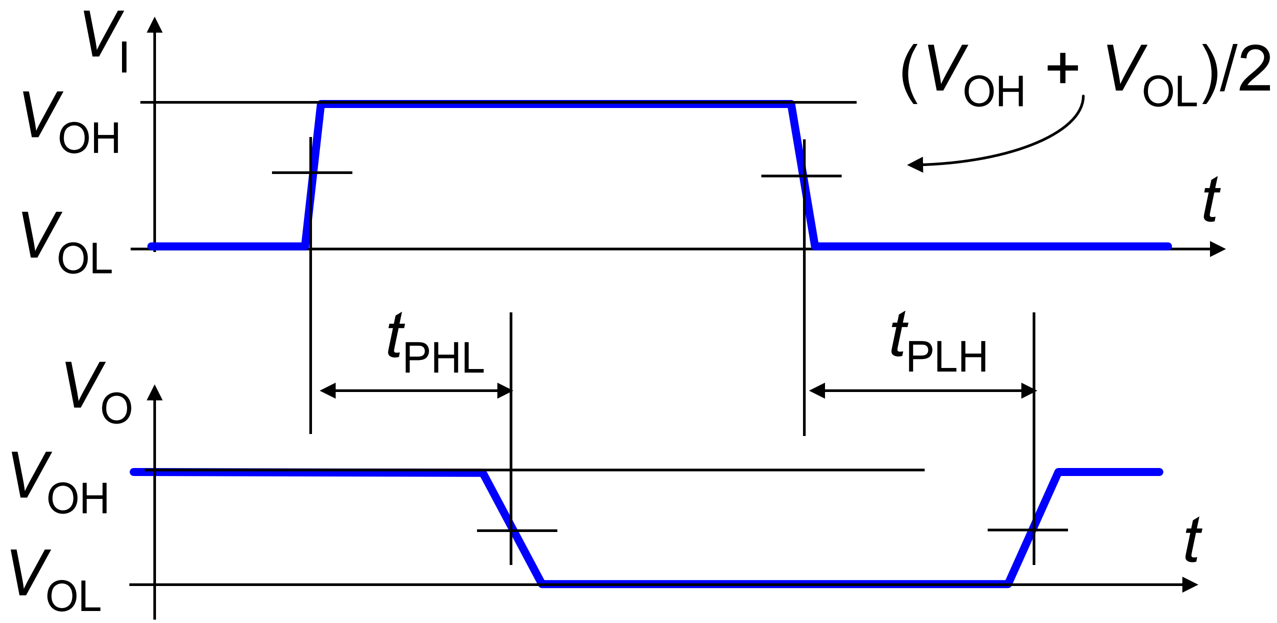
Rising and Falling Edges

- Rise time and fall time
  - They are defined at the 10 % and 90 % of the signal maximum variation



Delay Between Two Signals

- Delay between input and output signal (propagation time t_P)
  - At the 50 % of the whole range, from V_{OH} to V_{OL}

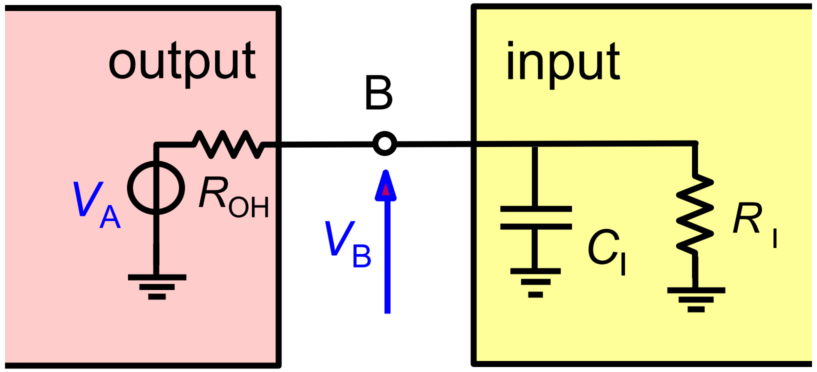


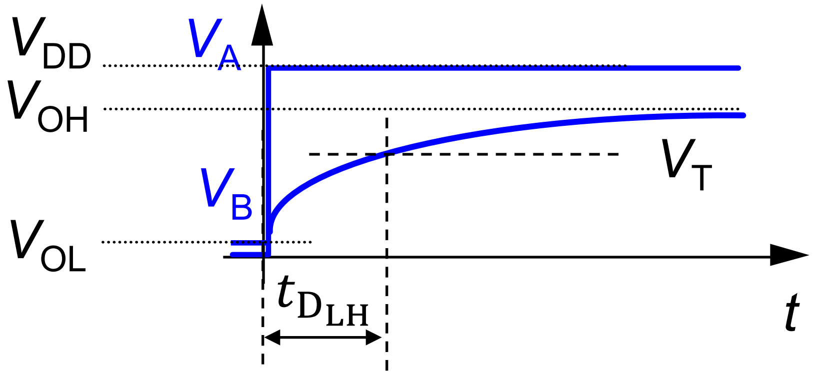
Delay RC Model: L→H Transition

Thevenin network (V_A, R_{OH}) for the output

R_I, C_I at input

The state change (L → H) is sensed when V_B crosses V_T → transmission delay, $t_{D_{LH}}$





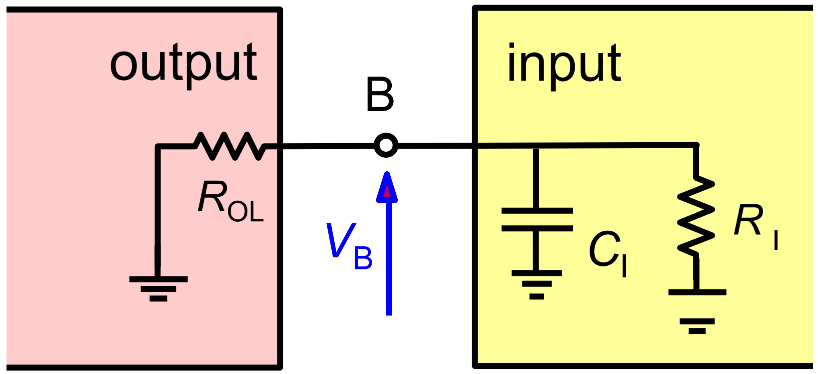
$$V_B(t) = (V_{initial} - V_{final})e^{-\frac{t}{RC}} + V_{final}$$

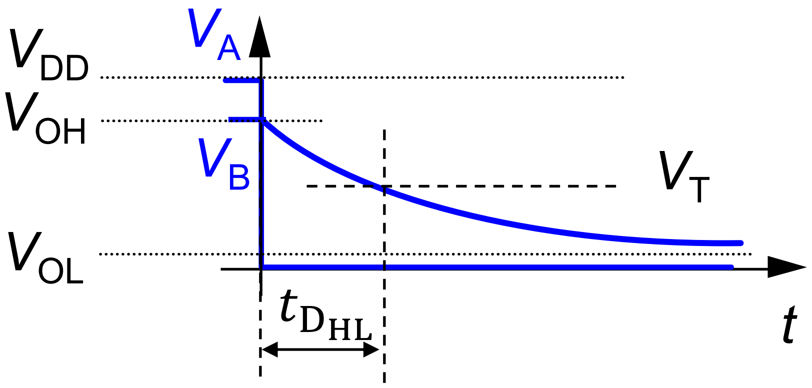
$$= (0\,V - V_A)e^{-\frac{t}{R_{OH}C_I}} + V_A$$

Thevenin network (V_A = 0 V, R_{OL}) for the output

R_I, C_I at input

The state change (H → L) is sensed when V_B crosses V_T → transmission delay, $t_{D_{HL}}$

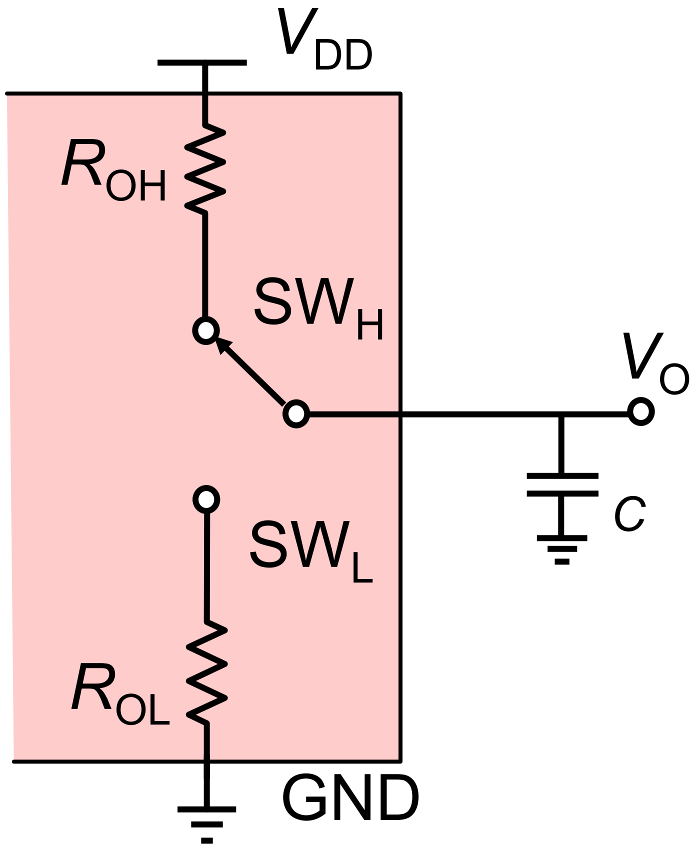




$$V_B(t) = (V_{\text{initial}} - V_{\text{final}})e^{-\frac{t}{RC}} + V_{\text{final}}$$
$$= (V_A - 0\text{ V})e^{-\frac{t}{R_{OL}C_I}} + 0\text{ V}$$

Model for CMOS Gate Delay

- The equivalent output resistance R_O is:
  - R_{OH} for state H
  - R_{OL} for state L
- By design, the values of R_{OL} and R_{OH} are similar

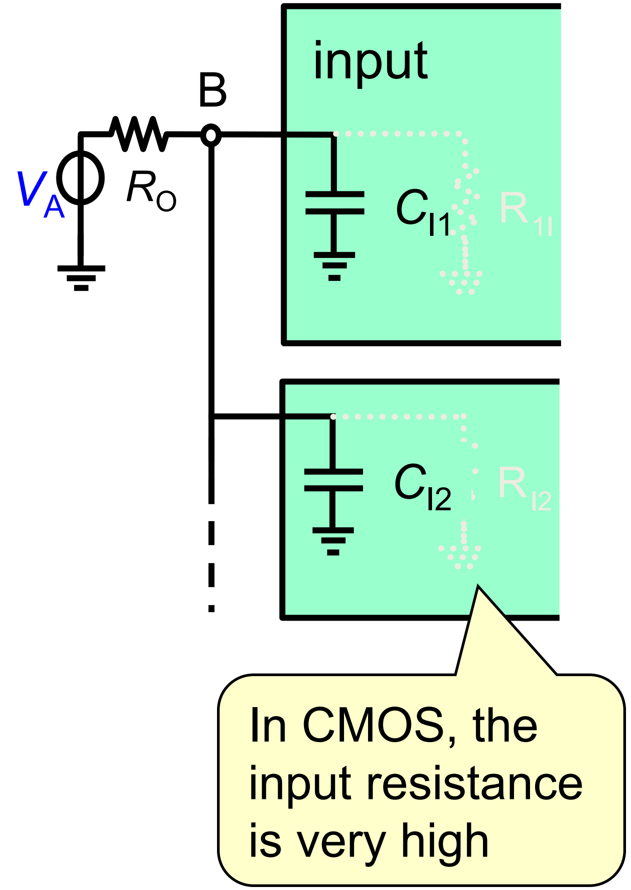


Effect of the Load

- The equivalent total capacitance, C_T, depends on # of driven inputs

$$C_T = \sum_i C_{I_i}$$

- More connected (driven) inputs
  - Delay and transition time increase
  - High time constant → slow transitions
- Max # of driven inputs by an output (fanout) is limited by *capacitive load*



Total Propagation Delay

- Other *R* and *C* inside the device (logic gates, …)
  - Cause additional delays input → output
- The delay (t_P) depends therefore on
  - Total *R* and *C* inside the device
  - Rising and falling edges of the input signals
  - Loads connected to the output
- On data sheets
  - Total maximum propagation delay
  - For specific measurement conditions
    - Load *C*, V_{DD}, input signal rising and falling edges

Logic Families

- Logic circuits are grouped into families
- Components of the same family have the same (or similar) electrical characteristics
- Components of the same family are statically compatible
  - Satisfy electrical compatibility conditions
    $V_{IL} > V_{OL}$, $V_{IH} < V_{OH}$

Characteristics of Logic Families

- Power supply voltage (5 V, 3.3 V, 2.5 V, …)
- Electrical parameters (input and output *V* and *I*)
- Delays and speed
- Power consumption
- TTL (now obsolete): low-power Schottky (LS),
  - Low-power fast Schottky (LS), prevent BJT saturation
  - Fast (F), higher currents for ultra-high speed (much faster than LS)
- CMOS families
  - High speed (HC): wide supply (2–6 V); advanced (AC): faster, > output current
  - Low supply voltage (LV/LVC): 3.3 V to reduce dynamic power consumption
  - Compatibility with TTL: HCT, ACT (tuned to TTL levels), BCT (BiCMOS with CMOS logic, BJT output for higher drive), LVT (3.3 V nominal, 5 V tolerated), …

Review Questions

- Why electronic systems are increasingly "digital“?
- List the static parameters of logic gates.
- What are the compatibility conditions between logic gates?
- What is the noise margin?
- How does the correct interfacing in a logical circuit occur?
- List the dynamic parameters of logic gates.
- What is the fanout and what does it depend on?